

Abubakar Sadiq Shittu

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Education

- Ph.D. Computer Science**, *University of Tennessee, Knoxville* Jan 2025 – Anticipated: Dec 2027
GPA: 3.67/4.00 | Coursework: Advanced Software Engineering, Algorithms, Data-Driven Security & Privacy, Computer Systems Organization, Digital Archaeology.
- M.Sc. Advanced Computer Science**, *Newcastle University, United Kingdom* Sept 2023 – Sept 2024
GPA: 4.00/4.00 (Distinction) | Coursework: Machine Learning, Secure Software Development, Systems Security, Risk & Trust Management.
- B.Sc. Computer Science**, *Fountain University, Osogbo* March 2018 – June 2021
GPA: 4.63/5.00 (First Class Honours)

Experience

- Graduate Research Assistant** Jan 2025 – Present
Usable Security and Engineering Research Laboratory, Led by [Dr. Scott Ruoti](#) *United States*
– Architected a high-throughput, containerized data ingestion pipeline using Docker and Celery for distributed task scheduling; integrated GitHub REST and GraphQL APIs with a PostgreSQL backend to continuously harvest 4M+ user profiles and 20M+ commit records [1].
– Spearheaded longitudinal mixed-methods studies on (1) TLS client authentication, triangulating participant feedback, server logs, and 400+ certificate signing requests to identify usability failures [2]; and (2) git commit signing, validating qualitative findings against .git log data and usability benchmarks (SUS/ASQ) to pinpoint technical and usability barriers to its adoption as a supply chain defense [manuscript in progress].
– Executing a interview study with developers to explore mental models and systemic barriers identified in prior large-scale data analyses, providing human-centric context to quantitative adoption trends [manuscript in progress].
- Technical and Administrative Assistant** Oct. 2021 – Oct. 2022
Kaduna State University *Nigeria*
– Served as the primary technical point of contact for the *examination and records unit*, diagnosing and resolving hardware, software, and network connectivity issues to ensure operational continuity.
- Information Technology Helpdesk Intern** Mar. 2019 – Aug. 2019
Nigerian National Petroleum Corporation *Nigeria*
– Supported 100+ computer users as the primary technical point of contact, resolving 5+ daily issues.

Preprints

- [1] **A. S. Shittu**, J. Sadik, F. Gholamrezae, and S. Ruoti, “Analysis of commit signing on GitHub,” *arXiv preprint arXiv:2604.14014*, 2026. [Online]. Available: <https://arxiv.org/pdf/2604.14014>.
- [2] **A. S. Shittu**, C. Shubert, J. Sadik, and S. Ruoti, “Understanding student experiences with TLS client authentication,” *arXiv preprint arXiv:2604.14330*, 2026. [Online]. Available: <https://arxiv.org/pdf/2604.14330>.

Selected Class Projects

- Machine Learning Pipelines**, *Newcastle University, UK* Dec 2023
– Built end-to-end pipelines for classification, regression, and NLP clustering; tuned Random Forest on FARS (100K+ records) improving ROC-AUC 88.05% → 93.09%; benchmarked SVM, Gradient Boosting (90.50%), and Logistic Regression (93.25%) with SMOTE; predicted bacterial growth parameters (μ , τ , a_0) across 19,992 experiments; compared TF-IDF vs. embeddings for K-Means clustering on 2,225 news articles; and tuned an LSTM on Raspberry Pi sensor data achieving $R^2 = 0.95$; produced visualizations and performance plots throughout using Matplotlib..
- Advanced RAG Pipeline** Spring 2025
– Built a RAG QA pipeline (LangChain + Gemini) with RecursiveCharacterTextSplitter chunking and VertexAIEmbeddings for retrieval; scraped and normalized a university course-schedule (BeautifulSoup) into a retrieval-optimized JSON dataset; indexed records as LangChain Documents in a vector store to serve natural-language queries.

Scholarships and Awards

- **Travel Grant**, *Summer School on Real-World Crypto and Privacy, Vodicce, Croatia* 2024
- **Chevening Scholarship**, *United Kingdom*. ~\$60,000: full tuition, monthly stipend, and travel 2023
- **Commonwealth Shared Scholarship** (Declined), *United Kingdom*. ~\$60,000 2023
- **100% Scholarship** (Declined) – *M.Sc. Computer Engineering, Kadir Has University, Turkey* 2022
- **Most Outstanding Student**, *College of Natural and Applied Sciences, Fountain University* 2019

Skills, Service, and Affiliations

- **Development:** Python, C/C++, Docker, SQL, PostgreSQL, REST/GraphQL APIs | **Systems & Analysis:** macOS, Windows, Linux/Unix, Bash, SonarQube, Wireshark, Git, NVivo, MAXQDA.
- **Reviewing:** [Journal of Cybersecurity](#), Q1 (SJR), 2025 | **Affiliations:** Graduate Student Member, IEEE & IEEE Computer Society | **Volunteering:** [Tennessee Extension Volunteer](#)